MIL-I-20565H
21 December 1977
SUPERSEDING
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22 June 1970

MILITARY SPECIFICATION

ICE CREAM CABINETS, MECHANICALLY REFRIGERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of self-contained, mechanically refrigerated, ice cream cabinet.

1.2 Classification. Ice cream cabinets covered by this specification shall be of the following minimum sizes as specified (see 6.2).

Size 1 - 15-gallon(57 L) capacity.
Size 2 - 25-gallon(94 L) capacity.
Size 3 - 35-gallon(132 L) capacity.
Size 4 - 45-gallon(170 L) capacity.
Size 5 - 70-gallon(265 L) capacity.
Size 6 - 80-gallon(303 L) capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants
LLL-B-810 - Building Board (Hard Board), Hard Pressed, Vegetable Fiber

MILITARY

MIL-R-12323 - Refrigerator and Related Equipment, Packaging and Packing Of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard No. 13 - Home Freezers, Methods of Rating and Testing

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

B 117 - Salt Spray (Fog) Testing.

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(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES INC. (UL)

Standard No. 471 - Commercial Refrigerators

(Application for copies should be addressed to Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; 333 Pfingsten Road, Northbrook, IL 60062; or 2602 Tampa East Blvd., Tampa, FL 33619.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 1 - Soda Fountain and Luncheonette Equipment

Seal of Approval Listing for Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The ice cream cabinets delivered under this specification shall be the manufacturer's current standard commercial product, except for any changes necessary to comply with specification requirements.

3.2 Standards.

3.2.1 UL standards. The ice cream cabinet shall conform to the applicable requirements of UL Standard No. 471.

3.2.1.1 UL certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the ice cream cabinets he proposes to furnish under this specification meet the requirements of UL Standard No. 471. Acceptable evidence of meeting the applicable requirements of UL standards shall be the UL label, a UL listing mark or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the ice cream cabinets offered have been tested and conform to the applicable requirements of UL Standard No. 471.

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3.2.2 NSF standard. Ice cream cabinets shall conform to the applicable requirements of NSF Standard No. 1.

3.2.2.1 NSF certification. The contractor shall submit satisfactory evidence to the contracting officer and his authorized representative that the ice cream cabinets he proposes to furnish under this specification meet the applicable requirements of NSF Standard No. 1 as follows:

(1) Listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished ice cream cabinet shall be submitted prior to approval of the first shipment, or

(2) A certification for the ice cream cabinet issued by NSF under their special one-time evaluation/certification service shall be submitted prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the cabinets are being procured, indicating that the ice cream cabinets have been tested and conform to NSF Standard No. 1 shall be submitted prior to commencing production.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for ice cream cabinets provided the completed item complies with all provisions of this specification (see 6.3).

3.3.1 Corrosion resisting sheet steel. Corrosion resisting steel sheet shall conform to the chemical composition of any class in the 300 series specified in ASTM A 167.

3.4 Design and construction. The ice cream cabinets shall consist of an insulated enclosure with lids located on the top to give access to a storage compartment where standard 2-1/2-(9 L), 3-(11 L), and 5-gallon(19 L) bulk ice cream containers with maximum diameters of 9-7/8 inches (251 mm) shall be stored. An automatic, self-contained refrigeration system shall be provided to maintain the selected storage compartment temperature (see 3.4.3.3).

3.4.1 Dimensions. The capacity is based on ice cream containers with maximum diameter of 9-7/8 inches (251 mm). Dimensions of each ice cream cabinet shall be as follows:

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TABLE I. Dimensions

Capacity (minimum)	Length (inches) (m) (maximum)	Width (inches) (m) ± 1 inch (± 25 mm)	Height (inches) (m) ± 1 inch (± 25 mm)
1. 15 gal.(57 L)	44(1.1)	31(0.8)	33-1/2(0.85)
2. 25 gal.(94 L)	53(1.3)	31(0.8)	33-1/2(0.85)
3. 35 gal.(132 L)	65(1.6)	31(0.8)	33-1/2(0.85)
4. 45 gal.(170 L)	78(1.9)	31(0.8)	33-1/2(0.85)
5. 70 gal.(265 L)	118(2.9)	31(0.8)	33-1/2(0.85)
6. 80 gal.(303 L)	81-1/2 (2)	31(0.8)	33-1/2(0.85)

3.4.2 Cabinet. The cabinet shall form a completely rigid structure to provide support to a full load of ice cream and the complete refrigerating system. Metal framing member, if used, shall be galvanized and phosphatized. To insure minimum heat transfer, there shall be no continuous metal connection between the outer shell and inner liner except as required by UL Standard No. 471.

3.4.2.1 Outer shell. The outer shell, except for the top, shall be fabricated of steel sheet having a minimum thickness of 0.0330 inch (0.8 mm). Joints shall be spot- or seam-welded where practicable. Joints that are not welded air-tight shall be sealed to prevent moisture infiltration into the insulation.

3.4.2.2 Inner liner. The sides of the inner liner shall be constructed from galvanized steel sheet having a nominal thickness of 0.0270 inch (0.7 mm). Galvanized steel sheet having a minimum thickness of 0.0827 inch (2 mm) shall be used at the bottom of the compartment. If the bottom of the liner is stiffened underneath by rigid insulation or marine plywood, the nominal thickness of the steel may be reduced to 0.0329 inch (0.8 mm). Seams shall be locked and soldered, or locked, spot-welded, and mastic sealed to provide a leak proof seam that is sealed and smooth.

3.4.2.3 Top. The top shall be one-piece construction of stainless steel sheet specified in 3.3.1 and have minimum thickness of 0.0220 inch (0.5 mm). Any rounded edges shall extend over the front and back of the cabinet only. The top shall be insulated and sealed and shall be reinforced.

3.4.2.4 Breaker strips. The breaker strips shall be fiberboard conforming to type II of LLL-B-810, styrene plastic, vinyl-chloride plastic, or a combination of these materials, except, when foamed-in-place polyurethane insulation with a fluorocarbon blowing agent is foamed directly against plastic

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breaker strips the plastic used shall be acrylonitrate-butadiene-styrene (ABS) plastic. The breaker strips shall join the inner liner and outer shell, make a vapor seal with the inner liner and outer shell, and make a smoothly finished appearance without recesses. The edges of the breaker strips shall be eased or rounded.

3.4.2.5 Lids. The lids shall be of flip-flop design (totally removable lid, hinged in the center to permit opening either half). The lids shall have a reflective surface of stainless steel sheet specified in 3.3.1 on the top and bottom side. Lifting handles shall be installed on the lids, and bumpers shall be provided to prevent marring of the top surface when the lids are open. Lid openings shall be of sufficient size and number to permit insertion of the total capacity of 2-1/2-(9L), 3-(11 L), or 5-gallon(19 L) standard ice cream containers. Every container in the top layer shall be accessible for direct dipping of ice cream when the lid(s) are open. Standard ice cream containers to be used with the cabinets shall have maximum outside diameters of 9-7/8 inches (251 mm). Gaskets for lids, if provided as part of the manufacturer's standard product, shall be as specified in 3.4.5.

3.4.2.6 Ice cream cans. When specified (see 6.2), ice cream cans of tinned steel sheet or stainless steel, in quantities with sizes indicated, shall be furnished.

3.4.2.7 Insulation. Insulation shall be industrial-type fiberglas insulation board, expanded polystyrene, or fluorocarbon expanded polyurethane preformed to shape or foamed-in-place. The insulation shall be homogeneous throughout, and shall have a "K" factor rating not greater than the specified values at a 75°F (24°C) mean temperature. The "K" mean factor shall not be more than 0.25 British thermal unit (Btu) per hour, per square foot, per degree temperature differential, per inch of thickness (0.04 W/m.k) for fiberglas insulation board and expanded polystyrene and not more than 0.15 (0.02 W/m.k) for expanded polyurethane. Loose fill type insulation shall not be acceptable. The lids and the spaces between the inner liner and outer shell shall be completely filled with any type insulation specified. Provisions shall be made to preclude moisture entering the insulation of ice cream cabinets having the evaporator or part of the evaporator bonded to the insulation side of the inner liner, or having tubing or other refrigerant containing parts subject to temperature below 32°F (0°C) in contact with the insulation.

3.4.2.8 Outer shell temperature. The temperature of the outer shell of the cabinets, including the top, shall not be more than 10°F (6°C) below the ambient temperature when the refrigeration system is in operation and the temperature in the storage compartment is being held at -10°F (-23°C) $\pm$ 2°F (1°C), when tested as specified in 4.3.3.

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3.4.3 Refrigeration system. The refrigeration system and component parts shall be completely installed and connected to make a fully automatic, self-contained, mechanically cooled ice cream cabinet ready for connection to an external electric power source. The refrigeration system shall consist of an electric motor driven condensing unit, a liquid control, evaporator, and necessary accessories including controls, motor starters, and protectors, all mounted and constructed to prevent vibration.

3.4.3.1 Condensing unit. The condensing units shall be air-cooled and shall be a hermetic type, thermally or mechanically sealed. The condensing units shall use one of the refrigerants specified in BB-F-1421. The compressor may be the oil-cooled type with its own evaporator and condenser system separate and apart from the condenser and evaporator for the product compartment of the cabinet.

3.4.3.2 Evaporator. The evaporator for the ice cream cabinets shall be copper tubing fastened to the insulation side of the inner liner or shall be embossed refrigerated dividers in the cabinet or a combination of both. A thermal-mastic or asphalt material shall be used to fillet the tubing on the liner for maximum efficiency. The tubing shall not be placed where it will cause frost accumulation on the breaker strips.

3.4.3.3 Temperature control. The ice cream cabinets shall be equipped with a thermostat to maintain temperature control. Unless otherwise specified (see 6.2) and within the pulldown requirements, the temperature control shall be adjustable to maintain the storage compartment temperature between -10°F (-23°C) and +15°F (-9°C) in any ambient between 50°F (10°C) and 100°F (38°C) (see 3.4.3.5). The thermostat shall be installed and adjusted to assure that the temperature will not vary more than 5°F (2°C) from the setting (see 4.3.2). The thermostat control knob shall be located in the machine compartment and mounted to provide for adjustment or change of a temperature setting.

3.4.3.4 Refrigerant and oil charge. The assembled refrigeration system shall be completely dehydrated and charged with the amount of refrigerant and oil necessary for operation.

3.4.3.5 Performance. The refrigeration system shall maintain the storage compartment temperature specified in 3.4.3.3. The refrigeration system, when installed as part of a complete ice cream cabinet, shall be capable of undergoing a no-load pulldown, a no-load and a storage load test as specified in 4.3.2. A temperature of 0°F (-18°C) shall be attained in not more than 6 hours in an ambient temperature of 110°F (43°C) for the no-load pulldown test. The running time for the no-load test shall not exceed 65 percent over a 6-hour period. The running time for the storage load test shall not exceed 70 percent over a 6-hour period. The ambient temperature for the no-load and storage load

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test shall be 90°F (32°C) and the stabilized freezer temperature at the beginning of the tests and during the tests shall be 0°F ± 2°F (-18° ± 1°C). Ice cream cabinets required for 50-cycle electric current operation shall meet the test temperature requirements when operating on a 50-cycle electric current at the motor manufacturer's rated motor voltage except that the running time shall not exceed 80 percent for the storage load test (see 4.3.2). Ice cream cabinets shall be tested with the current characteristics specified for the unit in the procurement document. Ice cream cabinets shall not cut out on overload during these no-load and storage load tests.

3.4.4 Electrical requirements. Unless otherwise specified (see 6.2), ice cream cabinets shall be designed for operation on a nominal 120-volt, 60-hertz, single-phase alternating current (a.c.) system. All ice cream cabinets shall be completely electrically wired. When permitted by UL Standard No. 471, a power supply cord shall be furnished as part of the ice cream cabinet. The cord shall have an attachment 3-prong plug with dead front construction and shall extend at least 5 feet (1.5 m), but not more than 9 feet (3 m) beyond the point at which it exits from the cabinet. Where power supply cords are prohibited by UL Standard No. 471, ice cream cabinets shall be provided with inclosed terminal blocks for connection to external circuits. The contractor may furnish a transformer to meet the specified electrical characteristics.

3.4.5 Gaskets. The gaskets shall be rubber, synthetic rubber, or vinyl-chloride plastics in accordance with the manufacturer's standard practice. The gaskets shall be replaceable cushion type, continuous one-piece units and shall be symmetrically installed to provide airtight sealing.

3.5 Finish.

3.5.1 Interior. Galvanized surfaces shall be restored by metal spraying where the coating has been fractured, flaked, peeled, or removed in forming, brazing, or welding. Exposed interior surfaces shall be additionally finished in white baked enamel or aluminum baked acrylic enamel when this is in accordance with the manufacturer's standard practice. Surfaces to be finished with baked enamel or aluminum baked acrylic enamel shall be phosphatized and primed.

3.5.2 Exterior. Exterior surfaces except stainless steel shall be finished with baked alkyd-amine enamel, baked acrylic enamel, or porcelain enamel. Surfaces to be finished with baked alkyd-amine enamel, or baked acrylic enamel shall be phosphatized. The color of the exterior surface, except the top, base, and trim, if any, shall be white. The trim shall be of a contrasting color. The finish shall be durable, smooth, continuous, nonabsorbent, and all discoloration or imperfections shall be removed. Exposed stainless steel surfaces shall have the manufacturer's standard commercial polish.

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3.5.3 Exposure to salt spray. There shall be no visible corrosion of either the interior or exterior surface of the ice cream cabinet following exposure to a salt spray test of 48 hours as specified in 4.3.4.

3.5.4 Structural soundness. The complete ice cream cabinet shall show no damage or permanent deformation of any part of the cabinet, base, or refrigeration system when tested as specified in 4.3.5.1 and 4.3.5.2.

3.6 Repair and maintenance. Provisions shall be made to allow for adjustment, service, and replacement of the condensing unit without disturbing or disassembling any major component or equipment. In addition, there shall be free access to inspect, or service, and adjust any other component or equipment, as required for proper operation of the ice cream cabinet. Removable access covers and plates shall be provided, as neccessary, to meet this requirement.

3.7 Operation. Ice cream cabinets, when tested as specified in 4.3.6, shall demonstrate that: (1) The refrigeration system is in satisfactory operating condition, (2) operates according to control settings, (3) mechanical parts are in working order, (4) the moisture content of the refrigerant at 100°F (38°C) does not exceed 10 parts per million (PPM) for R-12 and 45 PPM for R-22, and (5) there are no refrigerant leaks.

3.8 Service after testing. After testing, the ice cream cabinet shall reach equilibrium at room temperature. The interior shall be dried of any condensation and thoroughly aired and inspected. Required adjustment of temperature control (see 3.4.3.3), shall be made prior to final preparation for packing and packaging.

3.9 Marking for identification. Ice cream cabinets shall be marked with the following information on a nonferrous metal plate plate at least 0.04 inch (1 mm) thick.

1. Nomenclature
2. Size (see 1.2)
3. National Stock Number
4. Manufacturers Model, Type, and Catalog No.
5. Serial Number
6. Contract or Order Number
7. "U.S."
8. Other information required by UL Standard No. 471

The identification plate shall be fastened to the cabinet with rivets, bolts, or screws. All parts, assemblies, and subassemblies shall be marked in accordance with the UL Standards, as applicable.

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3.10 Workmanship. The ice cream cabinets shall have all metal forming done with dies or fixtures to insure uniformity of size and shape. All kinks, bends, and winds shall be removed from sheet metal or structural sections used in the fabrication of the ice cream cabinets. Shearing and chipping shall be finished on all portions of the work. All bolt and rivet holes shall be punched or drilled. Rivets, when used, shall be driven to completely fill the holes, with full-sized heads neatly finished and in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they shall not work loose under vibration. Optional equipment, accessories, and piping shall be fastened and braced to prevent damage or loosening during transit or operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the affected items and to the operation.

4.2.2.1 Visual examination of operations.

a. Phosphatizing and rustproofing of metal frames, when applicable (see 3.4.2).
b. Sealing of openings in cabinet shells against moisture infiltration (see 3.4.2.1 and 3.4.2.2).
c. Sealing and reinforcing of top (see 3.4.2.3.).
d. Installation of insulation (see 3.4.2.7).

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e. Filleting of tubing as applicable (see 3.4.3.2).
f. Charging of refrigeration system (see 3.4.3.4).
g. Phosphatizing and primer coating of interior and exterior surfaces when applicable (see 3.5.1 and 3.5.2).
h. Shearing and chipping operations (see 3.10).
i. Punching and drilling of rivet or bolt holes (see 3.10).
j. Rivet holes completely filled by rivets, when applicable (see 3.10).
k. Dies or fixtures are used for metal forming (see 3.10).
l. Bolts and nuts drawn tight (see 3.10).

4.2.3 End item inspection. The lot shall consist of all completely fabricated ice cream cabinets of the same size offered for inspection at one time. The sample unit shall be one completely fabricated ice cream cabinet.

4.2.3.1 Visual examination. Examination shall be made of the ice cream cabinets for the defects listed in table II. The inspection level shall be II with acceptable quality levels (AQLs) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Coating fractured, flaked, peeled or missing (galvanized surfaces).		X
	Colors not as specified		X
	Enamel missing (where required)	X	
	Enamel not smooth or continuous, or discolored.		X
	Exposed stainless steel surfaces not in accordance with manufacturer's standard practice.		X
Construction and workmanship (general)	Kinks, or bends, not removed.		X
	Component missing.	X	
Breaker strips	Edges not eased or rounded.		X
Lids	Lids not flip-flop design.	X	
	Lifting handles or bumpers missing.		X
Marking	Missing; incomplete; illegible; of improper size; sequence or method of application.		X

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4.2.3.2 Dimensional and coating thickness examination. An examination shall be made of the ice cream cabinets for dimensional and coating thickness defects. Any dimension or coating thickness not within specified requirements shall be classified a defect. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. Each ice cream cabinet shall be tested as specified in 4.3.6. Any nonconformance shall be cause for rejection of the cabinet. One production unit from the initial lot shall be tested as specified in 4.3.1 through 4.3.5. Any nonconformance shall be cause for rejection of the lot.

4.2.4 Code and standards compliance. Proof of compliance with the requirements of UL, as applicable to 3.2.1 and NSF, as applicable to 3.2.2, shall be submitted to the Government representative.

4.2.5 Inspection of packaging. An inspection shall be made to determine that preservation-packaging, packing, and marking conform to MIL-R-12323.

4.3 Tests.

4.3.1 Preparation for performance tests.

(a) The refrigeration system for the ice cream cabinet shall be charged, filled with oil and the unit given a run-in-operation period of 8 hours at room temperature. Controls shall be checked and the power supply voltage shall be plus or minus 5 volts of the voltage specified for operation. Other electrical characteristics shall be as specified.

(b) The placement of the ice cream cabinet in the test room shall be made to simulate a final location in a counter installation. The cabinet shall be backed up to within 6 inches (152 mm) of a wall space, the ends closed up (except for toe space) with a temporary partition and with a clear operating space in front of the cabinet.

(c) Air circulation in the test room shall maintain the specified uniformity of temperature, but the cabinet shall be shielded from forced air currents. Ambient temperatures shall be measured at points 1 foot (0.3 m) above the floor and 3 feet (0.9 m) above the cabinet at the center point on three sides. Ambient temperatures at each point shall be maintained within $\pm$ 2°F ($\pm$ 1°C) of the specified test conditions. The test conditions shall be maintained during stabilization periods as well as during test runs.

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(d) Ambient temperatures shall be measured with thermocouples, resistance thermometers, or glass thermometers. The temperature sensitive part of the temperature measuring device shall be in good thermal contact with a metallic mass having an equivalent total heat capacity of not less than 10 or more than 20 grams of water.

(e) In preparation for any operating test, the ice cream cabinet shall be operated for the period specified to obtain steady state conditions. Immediately following the preparatory test test period, the cabinet shall be operated for the performance test period during which the elements of performance are determined and recorded. Temperature observations shall be made continuously or at equal intervals of not greater than 15 minutes. Reported temperatures shall be the arithmetical average of all readings taken at equal time intervals.

(f) The ice cream cabinet shall be loaded with a simulated load corresponding to 100 percent of the manufacturer's rated load.

(g) The interior ice cream cabinet temperature shall be measured with not less than eight temperature measuring devices, equally distributed throughout the load for the storage load test, and equally distributed throughout the cabinet for no-load tests.

(h) The condensing unit of the ice cream cabinet shall be placed in operation when the ambient temperature, determined as the average of the temperature measurment devices on the outside of the cabinet is as specified. During the test period, observations shall be made that the interior ice cream cabinet temperatures, determined as the average of the temperature measuring devices located in the cabinet, shall be as specified. The maximum interior temperature of any single temperature measuring device in the cabinet, at the end of the test when the temperatures have stabilized, shall not exceed 0°F (-18°C) for the no-load pulldown or 5°F (-15°C) for the no-load and storage load tests.

4.3.2 Performance test. A no-load pulldown test, a no-load, and a storage load test shall be performed on the ice cream cabinets according to ASHRAE Standard No. 13 and the requirements of this specification to determine compliance with 3.3.3.5. The thermostat control shall be checked for compliance with 3.3.3.3 by operating the ice cream cabinet with a temperature control setting of 0°F (-18°C) in a steady state ambient condition of 50°F (10°C) and after storage compartment temperature has stabilized and held at the control setting temperature over a 2-hour period. The ambient temperature shall be raised at a constant rate to 110°F (43°C) over a 6-hour period. Refrigeration units procured for operation on a frequency of 50 hertz shall

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also be tested with full (storage) load by operating for 4 hours at the specified frequency and at 90 percent of the specified system voltage followed by an operating test of 4 hours at specified frequency and at 110 percent of specified system voltage. Inability of the refrigerator to maintain the specified temperature shall constitute failure of this test.

4.3.3 Outer shell temperature test. The refrigeration system of a complete ice cream cabinet shall be operated in an ambient temperature of 90°F (32°C) ± 2°F (± 1°C) until the temperature in the storage compartment stabilizes at -10°F (-23°C), ± 2°F (± 1°C). The temperature of the outer shell shall then be measured at five locations (front, back, top, and sides) to determine compliance with 3.4.2.8. Temperature difference greater than 10°F (5°C) between outer shell and ambient temperatures shall constitute failure of the test.

4.3.4 Salt spray test. The interior and exterior surfaces of the ice cream cabinet shall be subjected to a salt spray in accordance with ASTM B 117 to determine compliance with 3.5.3. Sample 6 inch (152 mm) by 12 inch (305 mm) panels, typical of the inner and outer shell materials, finish, and trim may be subjected to the salt spray test in lieu of the complete cabinet.

4.3.5 Structural soundness tests.

4.3.5.1 Sturdiness test. The complete ice cream cabinet shall be tilted on one side bottom edge so that the bottom makes an angle of 45 degrees with the horizontal to determine compliance with 3.5.4. The test shall be repeated with the opposite bottom edge bearing the load. Any damage or deformation shall constitute failure of this test.

4.3.5.2 Drop test. The complete ice cream cabinet shall be loaded to 50 percent of normal capacity by volume with suitable materials, not in liquid form, having an overall density of 35 pounds per cubic foot (560 kg/m^3) to determine compliance with 3.5.4. The material used for loading shall be nonrigid or shall consist of increments. The load shall be distributed uniformly over the floor of the liner. One side of the ice cream cabinet at a time shall then be tilted 3 inches (76 mm) off a level concrete floor and be allowed to fall freely in the following order: Left side, rear, right side, front. The ice cream cabinet shall be held in the tilted position for the drop test by a single support which can be withdrawn so the ice cream cabinet is not hampered in its free fall or caused to fall in a plane other than parallel to the specified direction of fall. Any damage or deformation shall constitute failure of this test.

4.3.6 Operation test. Each ice cream cabinet shall be operated long enough to cycle off at a normal control setting to determine compliance with 3.7. Controls shall be adjusted as required for normal operation. After completion of this

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portion of the test, and with the cabinet temperature at 70°F (21°C) ± 5°F (± 3°C), all refrigerant joints shall be tested for leaks to determine compliance with 3.7. A halide leak detector capable of detecting a leak of 0.5 ounce (0.01 kg) per year or less shall be used. Leaking joints shall be repaired and the ice cream cabinet shall be retested. Where the combination moisture and liquid indicator is used, it shall indicate "dry" within the limits specified during and upon completion of this operating test (see 3.7). When there is no moisture indicator in the system the ice cream cabinet shall be operated continuously for one hour with the lid(s) open and the cabinet empty. The cabinet shall then be disconnected from the electric supply and allowed to stand for one hour with the lid open. A sample of refrigerant vapor shall then be withdrawn from the low pressure side of the system and its moisture content determined by the electrical method employing hygroscopic film or in accordance with BB-F-1421 to determine compliance with the maximum allowable moisture content specified in 3.7. The contractor may use the Karl Fischer moisture determination test in lieu of testing as specified above. When the Karl Fischer test is used, the cabinet shall be operated for 24 hours at its coldest thermostat setting with the lid(s) open prior to the one hour shut-down and subsequent extraction of the refrigerant test sample. If the moisture content of the refrigerant exceeds the limits specified, drier replacements shall be made as required to bring the moisture content of the system within the allowable limits. Failure to pass the requirements outlined in 3.7 shall constitute failure of this test.

5. PACKAGING

5.1 Levels A, B, and C. Preservation-packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-R-12323 (see 6.2).

6. NOTES

6.1 Intended use. The ice cream cabinets covered by this specification are intended for use in stocking bulk or packaged ice cream in food dispensing establishments.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of ice cream cabinet required (see 1.2).
(c) Quantities and sizes of ice cream cans, if required (see 3.4.2.6).
(d) Limits of temperature control if other than specified (see 3.4.3.3).
(e) When electromagnetic compatibility is required (see 3.4.3.6).
(f) Electric power source characteristics if other than specified (see 3.4.4).
(g) Selection of applicable levels of preservation-packaging and packing (see 5.1).

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6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the valus stated in US customary units shall be controlling.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air force - 99

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-I-20565H - Ice Cream Cabinets, Mechanically Refrigerated

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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